

ASSIGNMENT NO.01

BT-401

(GENETIC RESOURCES AND CONSERVATION)

Question no.01:

Explain breeding and its types :

Breeding system:

Breeding is sexual reproduction that produces offspring, usually animals or plants. It can only occur between a male and a female animal or plant. Animal breeding is a segment of animal science that addresses the evaluation of the genetic value of domestic livestock. • A breed is a group of domestic animals with a homogeneous appearance, behavior, and other characteristics that distinguish it from other animals.

Types of breeding:

- **Pure breeding**
- **Out breeding**
- **In breeding**
- **Line breeding**
- **Cross breeding**

○ Pure breeding:

Pure-breeding is the mating of males and females of the same breed or type. • A purebred flock can be managed as a single flock because all animals are of the same breed. The goal of purebred production is to provide superior genetics to the commercial sheep industry.

○ Out breeding:

Outbreeding, the reverse case of inbreeding, means production of offspring through mating between individuals unrelated or distantly related by ancestral descent. Outbreeding can increase the genetic diversity and introduce elite alleles into the breeding population.

○ In breeding:

Inbreeding is technically defined as the mating of animals more closely related than the average relationship within the breed or population concerned. For practical purposes, if two mated individuals have no common ancestor within the last five or six generations, their progeny would be considered outbreeds.

O Line breeding:

Line breeding is a term commonly used to describe milder forms of inbreeding. Typically it involves arranging matings so that one or more relatives occur more than once in a pedigree, while avoiding close inbreeding.

O Cross breeding:

In animal breeding: Crossbreeding. Crossbreeding involves the mating of animals from two breeds. Normally, breeds are chosen that have complementary traits that will enhance the offsprings' economic value. An example is the crossbreeding of Yorkshire and Duroc breeds of pigs.

Question no 02:

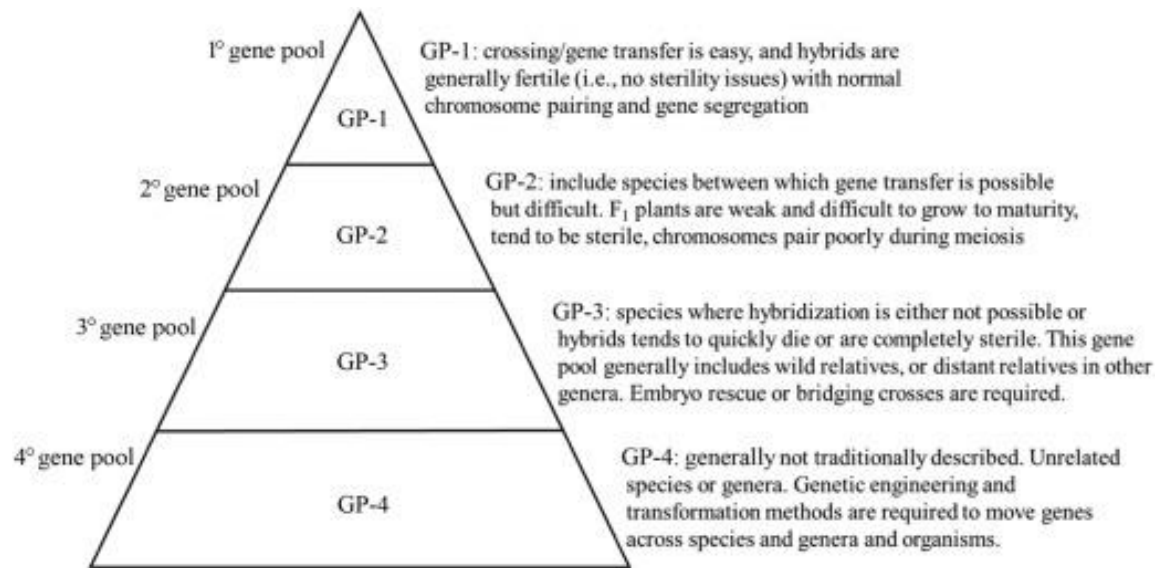
What is gene pool ? and discuss its types?

A gene pool is the total genetic diversity found within a population or a species. ... Inbreeding contributes to the creation of a small gene pool and makes populations or species more likely to go extinct when faced with some type of stress.

This can be any population in consideration for example frogs in a pond, trees in a forest, etc.

Types of gene pool:

- **Primary Gene Pool:** The members in this pool are of the same species and can inter-mate freely. ...
- **Secondary Gene Pool:** This gene pool consists of the species that are not the same but more closely related to a particular primary gene pool.

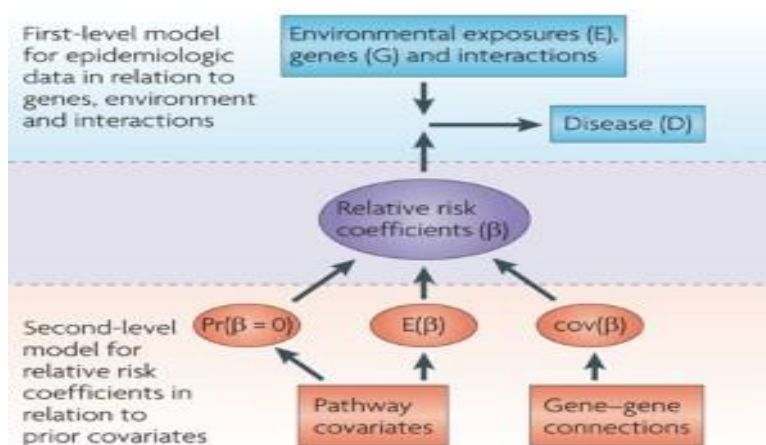


Question no. 03:

What is gene- environment interaction also discuss methods of its analysis?

Gene-environment interaction is defined as "a different effect of an environmental exposure on disease risk in persons with different genotypes," or, alternatively, "a different effect of a genotype on disease risk in persons with different environmental exposures. Not only do our genes and environment interact, as in range of reaction, but they also influence one another bidirectionally.

For example, the child of an NBA player would probably be exposed to basketball from an early age. Such exposure might allow the child to realize his or her full genetic, athletic potential.



Gene environment interaction is an influence on the expression of a trait that results from the interplay between genes and the environment. Some traits are strongly influenced by

genes, while other traits are strongly influenced by the environment. Most traits, however, are influenced by one or more genes interacting in complex ways with the environment.

Method of investigation:

The often used method to detect gene–environment interactions is by studying the effect a single gene variation (candidate gene) has with respect to a particular environment. Single nucleotide polymorphisms (SNP's) are compared with single binary exposure factors to determine any effects.

